

140X420mm



【Name and Strength of Active Ingredient(s)】

- RIVAMENSA PATCH 5：9mg as Rivastigmine
- RIVAMENSA PATCH 10：18mg as Rivastigmine

【Product Description】

- Rivamensa patch 5 (9mg as rivastigmine):
Transdermal patch of semi-transparent square shape with rounded corners in patch size of 5cm², with an adhesive film on one side
- Rivamensa patch 10 (18mg as rivastigmine):
Transdermal patch of semi-transparent square shape with rounded corners in patch size of 10cm², with an adhesive film on one side

【Pharmacodynamics】

Mechanism of Action：Pathological changes in dementia eg, Alzheimer's disease involve cholinergic neuronal pathways that project from the basal forebrain to the cerebral cortex and hippocampus. These pathways are known to be involved in attention, learning and memory, and other cognitive processes. Rivastigmine, a brain-selective acetyl- and butyryl-cholinesterase inhibitor of the carbamate type, is thought to facilitate cholinergic neurotransmission by slowing the degradation of acetylcholine released by functionally intact cholinergic neurons. Data from animal studies indicate that rivastigmine selectively increases the availability of acetylcholine in the cortex and hippocampus. Thus, Rivastigmine may have an ameliorative effect on cholinergic-mediated cognitive deficits associated with Alzheimer's disease and with Parkinson's disease. In addition, there is some evidence that cholinesterase inhibition could slow the formation of amyloidogenic β -amyloid-precursor protein (APP) fragments, and thus of amyloid plaques, which are one of the main pathological features of Alzheimer's disease.

【Pharmacokinetics】

Absorption：Absorption of rivastigmine from Rivastigmine patches is slow. After the first dose, detectable plasma concentrations are observed after a lag time of 0.5-1 hr. Concentrations then rise slowly and typically after 8 hrs reach levels close to maximum, although maximum values (Cmax) are often reached at later times (10-16 hrs). After the peak, plasma concentrations slowly decrease over the remainder of the 24-hr period of application. With multiple dosing (eg, at steady state), after the previous patch is replaced with a new one, plasma concentrations initially decrease slowly for about 40 min on average, until absorption from the newly applied patch becomes faster than the elimination and plasma levels begin to rise again to reach a new peak at approximately 8 hrs. At steady state, trough levels are approximately 50% of peak levels, in contrast to oral dosing, with which concentrations fall off to virtually zero between doses. This time course of plasma concentrations is observed with all patch strengths (sizes) in the investigated range of Rivastigmine Patch 5 to Rivastigmine Patch 20. Although less pronounced than with the oral formulation, exposure to rivastigmine (Cmax and AUC) increased over-proportionally with rising patch doses. Escalating from Rivastigmine Patch 5 to Rivastigmine Patch 20, the increase in rivastigmine AUC relative to the lowest dose of Rivastigmine Patch 5 was 2.6-, 4.9- and 7.8-fold for Rivastigmine Patch 10, Rivastigmine Patch 15 and Rivastigmine Patch 20, respectively. Patch has a much smaller fluctuation between trough and peak concentrations than for the oral formulation. Rivastigmine Patch 20 exhibited exposure (AUC24 hrs) in a typical patient equivalent to that which would be provided by an oral dose of about 9-10 mg twice daily (ie, 18-20 mg daily), while Rivastigmine Patch 10 exhibited exposure equivalent to that provided by an oral dose of about 6 mg twice daily (ie, 12 mg daily).

The inter-subject variability in rivastigmine pharmacokinetic parameters (normalised to dose/kg body weight) was lower than oral capsule after single dose or repeated doses.

A relationship between drug exposure at steady state (rivastigmine and metabolite NAP226-90) and body weight was observed in Alzheimer's dementia patients. Compared to a patient with a body weight of 65 kg, the rivastigmine steady-state concentrations in a patient with a body weight of 35 kg would be approximately doubled, while for a patient with a body weight of 100 kg the concentrations would be approximately halved. The effect of body weight on drug exposure suggests special attention to patients with very low body weight during up-titration (see Warning & Precaution).

Rivastigmine was well released from the transdermal system over a 24-hr dermal application with approximately 50% of the drug load released from the system.

Exposure (AUC ∞) to rivastigmine (and metabolite NAP266-90) was highest when the patch was applied to the upper back, chest or upper arm. Two other sites (abdomen and thigh) could be used if none of the 3 other sites is available, but the practitioner should keep in mind that the rivastigmine plasma exposure associated with these sites was approximately 20-30% lower.

There was no relevant accumulation of rivastigmine or the metabolite NAP226-90 in plasma in patients with Alzheimer's disease, except that with patch treatment plasma levels on the second day were higher than on the first.

Distribution：Rivastigmine is weakly bound to plasma proteins (approximately 40%). It readily crosses the blood-brain barrier and has an apparent volume of distribution in the range of 1.8-2.7 L/kg.

Metabolism：Rivastigmine is rapidly and extensively metabolized with an apparent elimination half-life in plasma of approximately 3.4 hrs after patch removal. Elimination was absorption rate limited (flip-flop kinetics), which explains the longer half-life after patch (3.4 hrs) versus oral or IV administrations (1.4-1.7 hrs). Metabolism is primarily via cholinesterase-mediated hydrolysis to the decarbamylated metabolite. In vitro, this metabolite shows minimal inhibition of acetylcholinesterase (<10%). The major cytochrome P-450 isoenzymes are minimally involved in rivastigmine metabolism. Total plasma clearance of rivastigmine is consistent with the nonlinear, overproportional pharmacokinetics of rivastigmine due to saturation of its elimination.

Less metabolism occurred after dermal treatment. Less NAP226-90 is formed following patch application, presumably because of the lack of presystemic (hepatic first pass) metabolism.

Elimination：Unchanged rivastigmine is found in trace amounts in the urine; renal excretion of the metabolites is the major route of elimination. Following administration of ¹⁴C-rivastigmine, renal elimination was rapid and essentially complete (90%) within 24 hrs. Less than 1% of the administered dose is excreted in the feces.

Special Populations：Elderly: Age had no impact on the exposure to rivastigmine in Alzheimer's disease patients treated with Rivastigmine patches.

Hepatic Impairment：No study was conducted with the Rivastigmine patches in subjects with hepatic impairment. After oral administration, the Cmax of rivastigmine was approximately 60% higher and the AUC of rivastigmine was more than twice as high in subjects with mild to moderate hepatic impairment than in healthy subjects.

Renal Impairment：No study was conducted with the Rivastigmine patches in subjects with renal impairment. After oral administration, Cmax and AUC of rivastigmine were more than twice as high in Alzheimer patients with moderate renal impairment compared with healthy subjects; however there were no changes in Cmax and AUC of rivastigmine in Alzheimer patients with severe renal impairment.

Toxicology：Acute Toxicity: The estimated oral LD50 values in mice were 5.6 mg base/kg (males) and 13.8-mg base/kg (females). The estimated oral LD50 values in rats were 8.1-mg base/kg (males) and 13.8-mg base/kg (females).

Repeated-Dose Toxicity：Oral and topical repeated-dose toxicity in animals revealed only effects associated with an exaggerated pharmacological action. No target organ toxicity was observed

Mutagenicity：Rivastigmine was not mutagenic in tests for gene mutation, primary DNA damage and chromosomal damage in vivo.

Carcinogenicity：No evidence of carcinogenicity was found in animals at the maximum tolerated dose. The exposure to rivastigmine and its metabolites was approximately equivalent to human exposure with highest doses of rivastigmine capsules and patches.

Teratogenicity：Reproductive Toxicity: In oral, no indication of teratogenic potential on the part of rivastigmine and no evidence of adverse effects of rivastigmine on fertility, reproductive performance or in utero or postnatal growth and development in rats.

Dermal Toxicity：Rivastigmine patches were not phototoxic. A mild irritant effect on the skin was observed. This may indicate a potential for Rivastigmine patches to induce mild erythema in patients.

【Indication】

- Treatment of patients with:
- Mild to moderately severe dementia of the Alzheimer's type,
 - Severe dementia of the Alzheimer's type
 - Mild to moderately severe dementia associated with Parkinson's disease.

【Recommended Dose】

Size(cm²)	API amount	Rivastigmine in vivo release rates per 24h
5	9mg	4.6mg
10	18mg	9.5mg

Initial Dose：Treatment is started with Rivastigmine Patch 5 once daily. After a minimum of 4 weeks of treatment and if well tolerated, this dose should be increased to Rivastigmine Patch 10, which is the recommended effective dose.

Maintenance Dose：Rivastigmine Patch 10 is the recommended maintenance daily dose which can be continued for as long as a therapeutic benefit for the patient exists. Individual responses to rivastigmine may vary and some patients may derive additional benefit from higher doses. Treatment should be temporarily interrupted if gastrointestinal adverse effects and/or worsening of existing extrapyramidal symptoms (eg, tremor) are observed until these adverse effects resolve. Patch treatment can be resumed at the same dose if treatment is not interrupted for more than several days. Otherwise treatment should be re-initiated with Rivastigmine Patch 5. If adverse effects persist on re-initiation of therapy, the dose should be temporarily reduced to the previous well-tolerated dose.

Switching from Capsules or Oral Solution：Patients treated with Rivastigmine capsules or oral solution may be switched to Rivastigmine patches as follows: A patient who is on a dose of oral rivastigmine (6 mg daily can be switched to Rivastigmine Patch 5. A patient who is on a stable and well tolerated dose of oral rivastigmine 9 mg/day can be switched to Rivastigmine Patch 10. If the oral dose of 9 mg daily has not been stable and well tolerated, a switch to Rivastigmine Patch

【Contraindication】

Known hypersensitivity to rivastigmine, other carbamate derivatives or to any of the excipients of Rivastigmine Patch (see Descriptions). Previous history of application site reactions suggestive of allergic contact dermatitis with rivastigmine transdermal patch (see Precautions).

【Warning and Precautions】

The incidence and severity of adverse events generally increase with increasing doses, particularly at dose changes. If treatment is interrupted for more than several days, it should be re-initiated with Rivastigmine Patch 5.

Gastrointestinal disorders eg, nausea and vomiting may occur when initiating treatment and/or increasing the dose. They may respond to a dose reduction. In other cases, use of Rivastigmine patches has been discontinued. Patients who show signs or symptoms of dehydration resulting from prolonged vomiting or diarrhoea can be managed with IV fluids and dose reduction or discontinuation if recognized and treated promptly. Dehydration can be associated with serious outcomes (see Adverse Reactions).

Patients with Alzheimer's disease may lose weight whilst taking cholinesterase inhibitors, including rivastigmine. The patient's weight should be monitored during therapy with Rivastigmine patches.

As with other cholinergic substances, care must be taken when prescribing Rivastigmine patches: To patients with sick sinus syndrome or conduction defects (sinoatrial block, atrioventricular block) (see Adverse Reactions); active gastric or duodenal ulcers or patients predisposed to these conditions because gastric acid secretions may be increased; patients predisposed to urinary obstruction and seizures because cholinomimetics may induce or exacerbate these diseases; history of asthma or obstructive pulmonary disease.

Like other cholinomimetics, rivastigmine may exacerbate extrapyramidal symptoms. In patients with dementia associated with Parkinson's disease who were treated with Rivastigmine capsules, worsening of parkinsonian symptoms, particularly tremor, has been observed.

Skin reactions : Skin application site reactions may occur with Rivastigmine patch and are usually mild or moderate in intensity. These reactions are not in themselves an indication of sensitization. However, use of rivastigmine patch may lead to allergic contact dermatitis.

Allergic contact dermatitis should be suspected if application site reactions spread beyond the patch size, if there is evidence of a more intense local reaction (eg, increasing erythema, edema, papules, vesicles) and if symptoms do not significantly improve within 48 hrs after patch removal. In these cases, treatment should be discontinued (see Contraindications).

In patients who develop application site reactions suggestive of allergic contact dermatitis to Rivastigmine patch and who still require rivastigmine, treatment should be switched to oral rivastigmine only after negative allergy testing and under close medical supervision. It is possible that some patients sensitized to rivastigmine by exposure to rivastigmine patch may not be able to take rivastigmine in any form.

There have been isolated post-marketing reports of patients experiencing disseminated skin hypersensitivity reactions when administered rivastigmine irrespective of the route of administration (oral, transdermal). In these cases, treatment should be discontinued (see Contraindications). Patients and caregivers should be instructed accordingly.

Special Populations : Patients with body weight <50kg may experience more adverse events and may be more likely to discontinue due to adverse events. Particular caution should be exercised in titrating these patients above the recommended maintenance dose of Rivastigmine Patch 10.

Hepatic Impairment : Patients with clinically significant hepatic impairment might experience more adverse events. Particular caution should be exercised in titrating these patients above the recommended maintenance dose of Rivastigmine Patch 10 (see Pharmacokinetics under Actions).

Effects on the Ability to Drive or Operate Machinery : Alzheimer's disease may cause gradual impairment of driving performance or compromise the ability to use machinery. Rivastigmine may induce dizziness and somnolence, mainly when initiating treatment or increasing the dose. Therefore, in patients with dementia treated with rivastigmine, the ability to continue driving or operating complex machines should be routinely evaluated by the treating physician.

Use in pregnancy : In animal studies, rivastigmine was not teratogenic. However, the safety of Rivastigmine in human pregnancy has not been established, and it should only be given to pregnant women if the potential benefit outweighs the potential risk for the fetus.

Use in lactation : It is not known if Rivastigmine is excreted into human milk, and patients on Rivastigmine should therefore not breastfeed.

QT Prolongation and torsade de pointes

Electrocardiogram QT prolongation may occur in patients treated with certain cholinesterase inhibitor products including rivastigmine. Rivastigmine may cause bradycardia which constitutes a risk factor in the occurrence of torsade de pointes, predominantly in patients with risk factors. Caution is advised in patients at higher risk of developing torsade de pointes: for example, those with uncompensated heart failure, recent myocardial infarction, bradyarrhythmias, hypokalemia or hypomagnesemia, personal or family history of QT prolongation, or concomitant use with medicinal products known to induce QT prolongation and/or torsade de pointes. Clinical monitoring may also be required.

【Interactions with other Medicaments】

No specific interaction studies have been conducted with Rivastigmine patches.

Rivastigmine is metabolized mainly through hydrolysis by esterases. Minimal metabolism occurs via the major cytochrome P-450 isoenzymes thus, no pharmacokinetic interactions are anticipated with other drugs metabolized by these enzymes.

No pharmacokinetic interaction was observed between rivastigmine and digoxin, warfarin, diazepam or fluoxetine in studies in healthy volunteers. The increase in prothrombin time induced by warfarin is not affected by administration of rivastigmine. No untoward effects on cardiac conduction were observed following concomitant administration of digoxin and rivastigmine.

Concomitant administration of rivastigmine with commonly prescribed medications eg, antacids, antiemetics, antidiabetics, centrally acting antihypertensives, β -blockers, calcium-channel blockers, inotropic drugs, antianginals, nonsteroidal anti-inflammatory drugs, oestrogens, analgesics, benzodiazepines and antihistamines, was not associated with an alteration in the kinetics of rivastigmine or an increased risk of clinically relevant untoward effects.

In view of its pharmacodynamic effects, rivastigmine should not be given concomitantly with other cholinomimetic drugs and might interfere with the activity of anticholinergic medications.

As a cholinesterase inhibitor, rivastigmine may exaggerate the effects of succinylcholine-type muscle relaxants during anaesthesia.

Medicinal products known to prolong the QT interval

Caution is advised when rivastigmine is used in combination with other medicinal products known to prolong the QT interval (including but not limited to quinidine, amiodarone, pimozide, halofantrine, cisapride, citalopram, mizolastin, moxifloxacin, erythromycin). Clinical monitoring may also be required.

Incompatibilities : To prevent interference with the adhesive properties of the patch, no cream, lotion or powder should be applied to the skin area where the Rivastigmine transdermal patch is to be applied.

【Pregnancy and Lactation】

Use in pregnancy : In animal studies, rivastigmine was not teratogenic. However, the safety of Rivastigmine in human pregnancy has not been established, and it should only be given to pregnant women if the potential benefit outweighs the potential risk for the fetus.

Use in lactation : It is not known if Rivastigmine is excreted into human milk, and patients on Rivastigmine should therefore not breastfeed.

【Side Effects】

The overall incidence of adverse events in patients treated with Rivastigmine Patch 10 was lower than the rate in patients who received Rivastigmine capsule treatment. Nausea and vomiting were the most common adverse events in patients who received active treatment and occurred at similar rates in both Rivastigmine Patch 20 and capsule groups. However, the rates of both of these events were substantially lower with Rivastigmine Patch 10 group. The most commonly reported adverse drug reactions are gastrointestinal including nausea and vomiting, especially during titration. Adverse reactions are ranked under headings of frequency, the most frequent first, using the following convention: Very common: common: uncommon: rare and very rare, including isolated reports.

Metabolism and Nutrition Disorders : Common : Anorexia, decreased appetite. Uncommon : Dehydration.

Psychiatric Disorders : Common : Anxiety, depression, insomnia. Uncommon : Agitation, delirium, hallucinations, aggression.

Nervous System Disorders : Common: Dizziness, headache. Uncommon: Cerebrovascular accident, syncope, somnolence, psychomotor hyperactivity.

Cardiac Disorders : Uncommon: Cardiac arrhythmia (eg, bradycardia, supraventricular extrasystole).

Gastrointestinal Disorders : Very Common: Nausea, Common: Vomiting, diarrhoea, dyspepsia, abdominal pain. Uncommon: Gastric ulcer, gastrointestinal haemorrhage (eg, haemorrhagic duodenitis).

Renal and Urinary Disorders : Uncommon: Urinary incontinence.

Skin and Subcutaneous Tissue Disorders : Uncommon: Hyperhidrosis.

General Disorders and Administration Site Conditions : Common: Application site reactions, erythema, pruritus and oedema: fatigue, asthenia. Uncommon: Contact dermatitis, malaise. Rare: Fall.

Investigations : Common: Decreased weight.

Infections and Infestations : Common: Urinary tract infection.

Additional adverse reactions observed in patients with dementia associated with Parkinson's disease treated with Rivastigmine transdermal patches: dehydration, weight decrease, aggression, visual hallucination (common). In patients with dementia associated Parkinson's disease the following adverse reactions have only been observed in clinical trials with Rivastigmine capsules: Nausea, vomiting (very common); decreased appetite, restlessness, worsening of Parkinson's disease, bradycardia, diarrhoea, dyspepsia, salivary hypersecretion, increased sweating (common); dystonia, atrial fibrillation, atrioventricular block (uncommon).

Additional Adverse Drug Reactions from Post-Marketing Spontaneous Reports</